Supplemental Material: The Geometry of Learning: Data Complexity Controls the Fractal Dimension of Gradient Descent

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PERTURBATION SENSITIVITY ANALYSIS

The perturbation scale ε used in Lyapunov exponent computation must be chosen carefully. We performed a sensitivity sweep over $\varepsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}, 10^{-9}\}$ at five learning rates for the baseline MLP (14K parameters, synthetic data, 5 seeds each).

At $\varepsilon = 10^{-8}$ and below, the measured Lyapunov exponent is uniformly $\lambda \approx +0.00004$ per training step regardless of learning rate—a numerical artifact arising from finite-precision arithmetic rather than genuine dynamical sensitivity. Converged measurements require $\varepsilon \geq 10^{-6}$. All results in the main text use $\varepsilon = 10^{-5}$.

This finding has implications for prior work: Lyapunov exponent measurements at $\varepsilon = 10^{-8}$ or smaller should be interpreted with caution, as the reported positive exponents may reflect numerical noise rather than chaotic dynamics.

CORRELATION DIMENSION PIPELINE CALIBRATION

We validated our D_2 pipeline against systems with known correlation dimensions, using the same trajectory length (∼400 points after transient removal) and algorithm parameters as the neural network experiments. Table I summarizes the results.

Two systematic biases are visible. Smooth manifolds are overestimated at higher dimensions (the 3-torus reads 3.74 instead of 3.0). Fractal attractors are underestimated, with bias growing with the true dimension, peaking around $D_2 \approx 5$ where the Mackey-Glass τ=30 system (expected 5.0) reads 3.17. This means our reported D_2 values for the CNN/CIFAR-10 attractor ($D_2 = 3.67$) are conservative lower bounds; the true fractal dimension is likely 15–35% higher.

TABLE I. D_2 pipeline calibration on known dynamical systems. Measured at $n = 400$ points, matching the production trajectory length used in the neural network experiments. Smooth manifolds are slightly overestimated; fractal attractors are underestimated, with bias growing with true dimension. Values are stable across $n \in \{200, 400, 800, 1600, 3200\}$ (see D_2(N) convergence analysis below).

System	Type	Expected D_2	Measured D_2
1-torus	smooth	1.0	1.02
2-torus	smooth	2.0	2.18
3-torus	smooth	3.0	3.73
4-torus	smooth	4.0	5.22
Hénon map	fractal	1.21	1.16
Rössler	fractal	1.99	1.72
Lorenz	fractal	2.05	1.73
Mackey-Glass τ=17	fractal	2.1	1.93
Mackey-Glass τ=23	fractal	3.6	2.44
Mackey-Glass τ=30	fractal	5.0	3.17
Mackey-Glass τ=50	fractal	7.0	6.03

EXTENDED D_2 DATA

Table II provides D_2 values across all learning rate fractions for the five experimental conditions.

Three features are visible. First, the CNN on CIFAR-10 peaks early (30% EoS) and declines toward $D_2 \approx 2$ at 95% EoS, while both MLPs on CIFAR-10 rise monotonically and peak at 90% EoS. Second, both synthetic conditions remain at $D_2 \approx 0.8$–1.0 across all learning rates, with standard deviations below 0.04. Third, cross-seed variance is largest at the transition: the CNN at 20% EoS shows $D_2 = 2.63 \pm 0.62$ and the MLP (269K) at 60% EoS shows $D_2 = 2.70 \pm 0.41$, reflecting sensitive dependence on initial conditions near the onset of multi-dimensional dynamics. With N=10 seeds, the MLP variances have widened from the suspiciously tight values seen at N=3 (e.g., MLP 156K at 90% EoS: ±0.10 versus the earlier ±0.02), confirming that the original error bars underestimated true seed-to-seed variability.

PERSISTENT HOMOLOGY DETAILS

Tables III–**??** report persistent homology feature counts for all three CIFAR-10 conditions. Persistence is computed via RIPSER on PCA-reduced (10-dimensional) function-space trajectories, subsampled to 400 points.

At 5% of EoS, the trajectory is a simple convergence path with zero H_1 features in all architectures. As the learning rate increases, hundreds of loops (H_1) and voids (H_2) appear at all persistence scales. The gap ratio remains near 1.0 throughout—no dominant topological feature emerges—which is the signature of fractal struc-

TABLE II. Correlation dimension D_2 (mean ± standard deviation across seeds) for all conditions at each tested learning rate fraction. Learning rates expressed as fractions of $2/\lambda_{\max}$.

% EoS	CNN+C-10 (269K, 10)	MLP+C-10 (269K, 10)	MLP+C-10 (156K, 10)	CNN+synth (269K, 10)	MLP+synth (14K, 20)
5	0.99 ± 0.005	0.97 ± 0.002	0.97 ± 0.001	0.94 ± 0.004	0.93 ± 0.006
10	1.03 ± 0.02	0.98 ± 0.001	0.98 ± 0.001	0.84 ± 0.01	—
15	1.29 ± 0.12	0.98 ± 0.002	0.98 ± 0.001	0.86 ± 0.02	—
20	2.63 ± 0.62	0.98 ± 0.002	0.98 ± 0.001	0.91 ± 0.02	0.90 ± 0.002
30	$\mathbf{3.67 \pm 0.08}$	0.98 ± 0.001	0.97 ± 0.008	0.95 ± 0.004	0.91 ± 0.001
40	3.46 ± 0.09	1.09 ± 0.03	1.13 ± 0.06	0.95 ± 0.01	—
50	3.10 ± 0.11	1.44 ± 0.10	1.74 ± 0.28	0.95 ± 0.01	—
60	2.72 ± 0.10	2.70 ± 0.41	3.24 ± 0.35	0.96 ± 0.01	—
70	2.42 ± 0.07	3.90 ± 0.17	4.08 ± 0.12	0.97 ± 0.04	—
80	2.15 ± 0.06	4.28 ± 0.07	4.40 ± 0.11	0.98 ± 0.03	—
90	2.00 ± 0.06	$\mathbf{4.37 \pm 0.05}$	$\mathbf{4.61 \pm 0.10}$	0.98 ± 0.03	—
95	1.92 ± 0.04	—	—	—	—

TABLE III. Persistent homology feature counts (mean across 3 seeds) for CNN/CIFAR-10. Gap ratio is the ratio of the longest to second-longest H_1 lifetime.

% EoS	H_1 features	H_2 features	H_1 gap ratio
5	0	0	—
15	107	50	1.24
20	233	173	1.11
30	386	355	1.16
40	421	401	1.07
60	378	305	1.05
90	299	163	1.07

TABLE IV. Persistent homology feature counts (mean across 3 seeds) for MLP/CIFAR-10, 156K parameters (width 50).

% EoS	H_1 features	H_2 features	H_1 gap ratio
5–25	0	0	—
30	1	0	—
40	77	17	1.24
50	181	92	1.25
60	255	163	1.09
70	313	233	1.09
80	360	308	1.01
90	389	337	1.06

ture rather than a smooth torus. A clean 2-torus would show exactly 2 dominant H_1 features with a gap ratio $\gg 1$, clearly distinguishable from the remaining topological noise.

The MLP conditions confirm that persistent homology tracks D_2 across architectures. Both MLPs show zero H_1 features below 30% EoS (where $D_2 \approx 1$), onset of topological complexity at 40% EoS (where D_2 first exceeds 1.0), and hundreds of features at 90% EoS (where $D_2 > 4$). The topological transition occurs at the same learning-rate threshold as the geometric transition measured by correlation dimension. At 90% EoS, both MLPs show comparable feature counts to the CNN at its peak (30% EoS: 386 H_1, 355 H_2), consistent with both architectures accessing the same attractor complexity when driven by the same data.

TABLE V. Persistent homology feature counts (mean across 3 seeds) for MLP/CIFAR-10, 269K parameters (width 85).

% EoS	H_1 features	H_2 features	H_1 gap ratio
5–30	0	0	—
40	42	4	1.10
50	137	57	1.14
60	217	132	1.11
70	307	240	1.10
80	358	297	1.34
90	392	315	1.11

D_2 CONVERGENCE WITH TRAJECTORY LENGTH

Figure **??** shows the measured D_2 as a function of trajectory length N for three neural network conditions alongside two reference systems (Lorenz attractor, Mackey-Glass τ=30). Longer trajectories were obtained by recording function-space outputs at every training step rather than every 10th step, within the standard 5,000-step protocol; this yields ∼4,000 post-transient points without altering the dynamics.

The reference systems confirm that the Grassberger-Procaccia pipeline produces stable D_2 estimates across

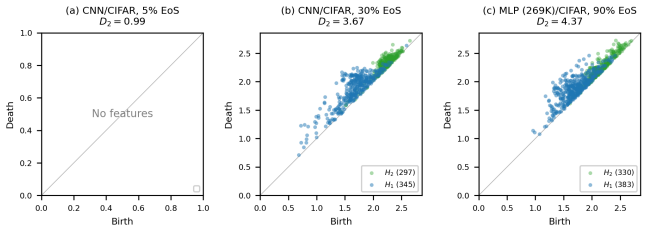

FIG. 1. Representative persistence diagrams. CNN/CIFAR-10 at 5% EoS (trivial: zero features), CNN/CIFAR-10 at 30% EoS (strange attractor: features at all scales), MLP/CIFAR-10 (269K) at 90% EoS (strange attractor: comparable topological complexity to CNN despite different architecture). Points near the diagonal are short-lived; points far from it are persistent topological features.

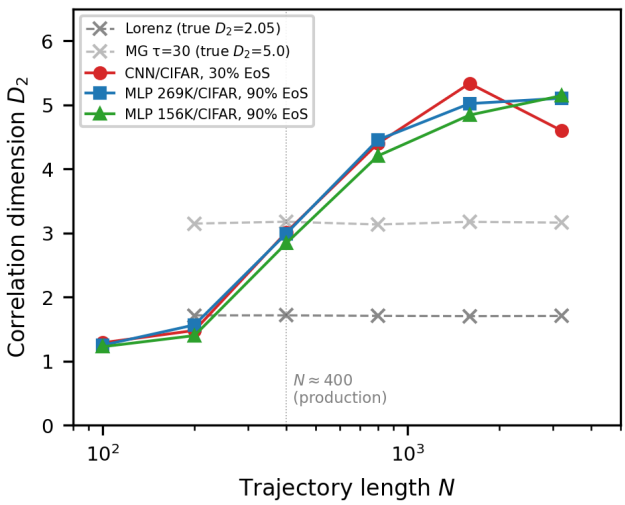

FIG. 2. D_2 versus trajectory length N. Reference systems (gray) are flat: Lorenz at $D_2 \approx 1.72$ and MG τ=30 at $D_2 \approx 3.17$ across all N, confirming pipeline stability. Neural conditions rise from N=100 to N=800 and approach plateau by N=1600–3200. Both MLPs change by $< 7\%$ from N=1600 to N=3200, stabilizing near $D_2 \approx 5.1$. The dashed line marks $N \approx 400$, the production trajectory length; reported D_2 values are conservative lower bounds.

the full range $N \in [200, 3200]$: the Lorenz attractor varies by $< 1\%$ and Mackey-Glass τ=30 by $< 2\%$. For the neural conditions, D_2 rises from N=100 to N=800 as more attractor structure becomes resolvable, then approaches a plateau. Both MLPs stabilize near $D_2 \approx 5.1$ (1.7% and 6.2% change from N=1600 to N=3200). The CNN shows more variation at long N (one seed), but the key result is that D_2 does not grow unboundedly—the attractor has a finite dimension. At the production length $N \approx 400$, all three conditions are on the rising portion of the curve, confirming that reported values are conservative lower bounds on the true attractor dimension.

CONVERGENCE OF D_2 AND λ WITH MEASUREMENT PARAMETERS

We verified that the choices $\varepsilon = 10^{-5}$ and $N_{\text{inputs}} = 100$ used in the main text are within converged regimes. For the CNN on CIFAR-10 at 30% of EoS (seed 0), we swept $N_{\text{inputs}} \in \{25, 50, 100, 200, 400\}$ and $\varepsilon \in \{10^{-4}, 5\times10^{-5}, 10^{-5}, 5\times10^{-6}, 10^{-6}\}$.

The correlation dimension is stable: D_2 varies by only 0.10 across all N_{inputs} values (range 3.74–3.83) and is completely independent of ε, as expected since ε affects only the perturbation protocol, not the trajectory. The production value $D_2 = 3.74$ at $N_{\text{inputs}} = 100$ sits in the center of this range.

The Lyapunov exponent shows more structure: λ varies by a factor of $\sim$3 across ε values, with a dip at $\varepsilon = 10^{-5}$. All values remain positive ($\lambda \in [0.8, 2.3]\times10^{-4}$ per step), confirming chaos across the full ε range tested. The variation with ε is consistent across all N_{inputs} values, indicating a genuine scale-dependent feature of the dynamics rather than a numerical artifact. Crucially, the qualitative conclusion—positive Lyapunov exponent at 30% of EoS—is robust to the choice of ε within $[10^{-6}, 10^{-4}]$.

DEPTH SCALING

We tested MLP depth scaling with tanh networks of width 50 on synthetic data (Table IV).

TABLE VI. Chaos window properties versus MLP depth. D_2 remains below 1.0 for all depths on synthetic data, but the chaos window widens with depth as Ruelle-Takens predicts. Peak λ is in units of inverse training steps, matching the main-text convention.

Depth	Params	Window width	Peak λ	Max PC2%
2	14K	0.038	$+1.35 \times 10^{-4}$	5.2
3	17K	0.046	$+2.36 \times 10^{-4}$	8.7
4	19K	0.145	$+3.12 \times 10^{-4}$	12.6
5	22K	0.137	$+2.36 \times 10^{-4}$	12.9

At depth 4 and 5, chaos persists across the entire tested learning rate range—the "basin wins" regime from the 2-layer MLP disappears. Off-axis dynamics grow to 13% of variance. However, D_2 remains below 1 at all depths on synthetic data, confirming that architectural depth alone is insufficient for multi-dimensional chaos without complex data.

LABEL-NOISE SWEEP

To test whether data complexity acts as a continuous control parameter rather than a binary switch, we

sweep label-noise fraction p from 0 (clean CIFAR-10) to 1 (uniformly random labels) at each architecture's peak-D_2 learning rate. A deterministic random number generator (seed 137) selects which samples are relabeled, so the noise pattern is identical across training seeds. The EoS threshold is measured once on clean data ($p = 0$) and held fixed. Three seeds per noise level.

TABLE VII. D_2 versus label-noise fraction p for two conditions. CNN at 30% EoS: D_2 falls as label structure is degraded. MLP 269K at 90% EoS: D_2 rises modestly as random labels roughen the loss surface.

	CNN/CIFAR (30% EoS)		MLP 269K (90% EoS)	
p	D_2	σ	D_2	σ
0.00	3.64	0.11	4.36	0.05
0.10	3.59	0.15	4.46	0.02
0.25	3.40	0.21	4.54	0.06
0.50	2.67	0.68	4.69	0.07
0.75	1.91	0.57	4.72	0.02
0.90	1.64	0.48	4.75	0.03
1.00	1.68	0.54	4.77	0.08

The CNN result is straightforward: structured labels create the oscillatory modes that produce multi-dimensional chaos; randomizing them removes these modes and collapses the attractor toward $D_2 \approx 1$, matching the CNN/synthetic baseline. The CNN variance also grows with noise (σ rises from 0.11 to 0.54), consistent with individual seeds responding heterogeneously as label structure is degraded.

The MLP result is subtler. At 90% of EoS, the dynamics are already near the stability boundary. Random labels create a rougher loss surface—more competing local minima, steeper curvature ridges—that sustains high-dimensional chaos even in the absence of label structure. The rise is modest ($\Delta D_2 \approx 0.4$, less than 10σ of the clean measurement) but monotonic across all seven noise levels.

This regime dependence is interpretatively important: "data complexity controls D_2" is accurate at moderate learning rates where the data creates the modes. Near the EoS, the relationship is more nuanced—the landscape roughness produced by incoherent labels provides an alternative source of high-dimensional dynamics.

λ–D_2 DISSOCIATION ACROSS ARCHITECTURES

The main text reports that peak Lyapunov exponent λ and peak correlation dimension D_2 occur at different learning-rate fractions—a dissociation characteristic of the KAM transition. Figure **??** shows the full $\lambda(\eta)$ and $D_2(\eta)$ curves for all three CIFAR-10 conditions, confirming that this dissociation is universal.

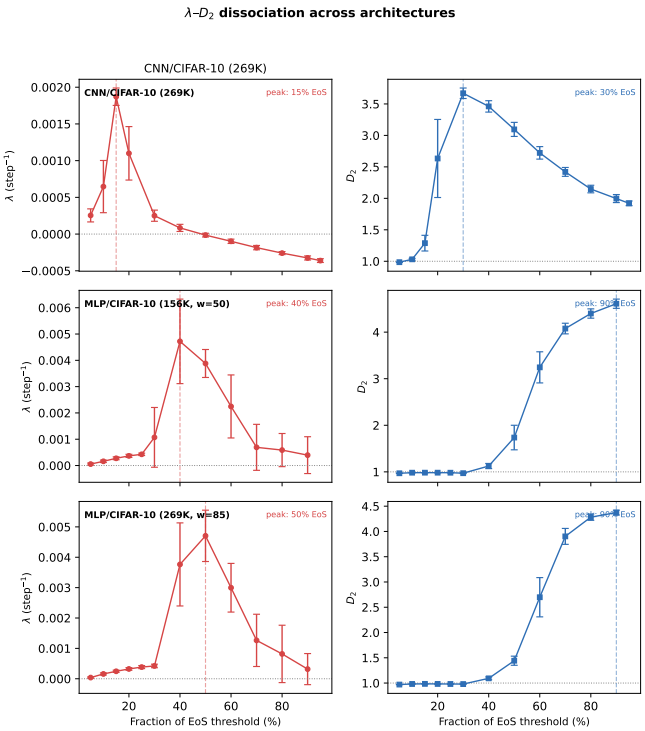

FIG. 3. λ (left) and D_2 (right) versus fraction of the EoS threshold for all three CIFAR-10 conditions. Dashed vertical lines mark each quantity's peak. In every case, peak λ precedes peak D_2: CNN at 15% vs 30%, MLP 156K at 40% vs 90%, MLP 269K at 50% vs 90%. Error bars: standard deviation across 10 seeds.

The dissociation gap varies with architecture: 15 percentage points for the CNN, 50 for the MLP 156K, and 40 for the MLP 269K. In each case, geometric complexity peaks not where perturbation sensitivity is greatest but in the transition zone where chaotic and basin-convergent dynamics compete. This is consistent with KAM theory, where the richest fractal structure emerges at the boundary between ordered and chaotic phase-space regions.